

Linear and Quadratic Discriminant Analysis: Tutorial

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Abstract

This tutorial explains Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) as two fundamental classification methods in statistical and probabilistic learning. We start with the optimization of decision boundary on which the posteriors are equal. Then, LDA and QDA are derived for binary and multiple classes. The estimation of parameters in LDA and QDA are also covered. Then, we explain how LDA and QDA are related to metric learning, kernel principal component analysis, Mahalanobis distance, logistic regression, Bayes optimal classifier, Gaussian naive Bayes, and likelihood ratio test. We also prove that LDA and Fisher discriminant analysis are equivalent. We finally clarify some of the theoretical concepts with simulations we provide.

1. Introduction

Assume we have a dataset of *instances* $\{(x_i, y_i)\}_{i=1}^n$ with sample size n and dimensionality $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. The y_i 's are the class labels. We would like to *classify* the space of data using these instances. Linear Discriminant Analysis (LDA) and Quadratic discriminant Analysis (QDA) (Friedman et al., 2009) are two well-known *supervised classification* methods in statistical and probabilistic learning. This paper is a tutorial for these two classifiers where the theory for binary and multi-class classification are detailed. Then, relations of LDA and QDA to metric learning, kernel Principal Component Analysis (PCA), Fisher Discriminant Analysis (FDA), logistic regression, Bayes optimal classifier, Gaussian naive Bayes, and Likelihood Ratio Test (LRT) are explained for better understanding of these two

fundamental methods. Finally, some experiments on synthetic datasets are reported and analyzed for illustration.

2. Optimization for the Boundary of Classes

First suppose the data is one dimensional, $x \in \mathbb{R}$. Assume we have two classes with the Cumulative Distribution Functions (CDF) $F_1(x)$ and $F_2(x)$, respectively. Let the Probability Density Functions (PDF) of these CDFs be:

$$f_1(x) = \frac{\partial F_1(x)}{\partial x}, \quad (1)$$

$$f_2(x) = \frac{\partial F_2(x)}{\partial x}, \quad (2)$$

respectively.

We assume that the two classes have normal (Gaussian) distribution which is the most common and default distribution in the real-world applications. The mean of one of the two classes is greater than the other one; we assume $\mu_1 < \mu_2$. An instance $x \in \mathbb{R}$ belongs to one of these two classes:

$$x \sim \begin{cases} \mathcal{N}(\mu_1, \sigma_1^2), & \text{if } x \in \mathcal{C}_1, \\ \mathcal{N}(\mu_2, \sigma_2^2), & \text{if } x \in \mathcal{C}_2, \end{cases} \quad (3)$$

where $\mathcal{C}_1$ and $\mathcal{C}_2$ denote the first and second class, respectively.

For an instance x , we may have an error in estimation of the class it belongs to. At a point, which we denote by x^* , the probability of the two classes are equal; therefore, the point x^* is on the boundary of the two classes. As we have $\mu_1 < \mu_2$, we can say $\mu_1 < x^* < \mu_2$ as shown in Fig. 1. Therefore, if $x < x^*$ or $x > x^*$ the instance x belongs to the first and second class, respectively. Hence, estimating $x < x^*$ or $x > x^*$ to respectively belong to the second and first class is an error in estimation of the class. This probability of the error can be stated as:

$$\mathbb{P}(\text{error}) = \mathbb{P}(x > x^*, x \in \mathcal{C}_1) + \mathbb{P}(x < x^*, x \in \mathcal{C}_2). \quad (4)$$

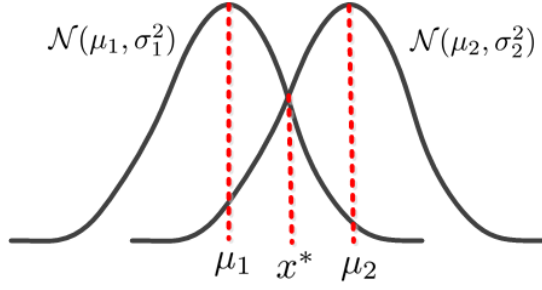


Figure 1. Two Gaussian density functions where they are equal at the point x^* .

As we have $\mathbb{P}(A, B) = \mathbb{P}(A|B) \mathbb{P}(B)$, we can say:

$$\begin{aligned} \mathbb{P}(\text{error}) &= \mathbb{P}(x > x^* | x \in \mathcal{C}_1) \mathbb{P}(x \in \mathcal{C}_1) \\ &\quad + \mathbb{P}(x < x^* | x \in \mathcal{C}_2) \mathbb{P}(x \in \mathcal{C}_2), \end{aligned} \quad (5)$$

which we want to minimize:

$$\underset{x^*}{\text{minimize}} \quad \mathbb{P}(\text{error}), \quad (6)$$

by finding the best boundary of classes, i.e., x^* .

According to the definition of CDF, we have:

$$\begin{aligned} \mathbb{P}(x < c, x \in \mathcal{C}_1) &= F_1(c), \\ \implies \mathbb{P}(x > x^*, x \in \mathcal{C}_1) &= 1 - F_1(x^*), \end{aligned} \quad (7)$$

$$\mathbb{P}(x < x^*, x \in \mathcal{C}_2) = F_2(x^*). \quad (8)$$

According to the definition of PDF, we have:

$$\mathbb{P}(x \in \mathcal{C}_1) = f_1(x) = \pi_1, \quad (9)$$

$$\mathbb{P}(x \in \mathcal{C}_2) = f_2(x) = \pi_2, \quad (10)$$

where we denote the priors $f_1(x)$ and $f_2(x)$ by π_1 and π_2 , respectively.

Hence, Eqs. (5) and (6) become:

$$\underset{x^*}{\text{minimize}} \quad (1 - F_1(x^*)) \pi_1 + F_2(x^*) \pi_2. \quad (11)$$

We take derivative for the sake of minimization:

$$\begin{aligned} \frac{\partial \mathbb{P}(\text{error})}{\partial x^*} &= -f_1(x^*) \pi_1 + f_2(x^*) \pi_2 \stackrel{\text{set}}{=} 0, \\ \implies f_1(x^*) \pi_1 &= f_2(x^*) \pi_2. \end{aligned} \quad (12)$$

Another way to obtain this expression is equating the posterior probabilities to have the equation of the boundary of classes:

$$\mathbb{P}(x \in \mathcal{C}_1 | X = x) \stackrel{\text{set}}{=} \mathbb{P}(x \in \mathcal{C}_2 | X = x). \quad (13)$$

According to Bayes rule, the *posterior* is:

$$\begin{aligned} \mathbb{P}(x \in \mathcal{C}_1 | X = x) &= \frac{\mathbb{P}(X = x | x \in \mathcal{C}_1) \mathbb{P}(x \in \mathcal{C}_1)}{\mathbb{P}(X = x)} \\ &= \frac{f_1(x) \pi_1}{\sum_{k=1}^{|\mathcal{C}|} \mathbb{P}(X = x | x \in \mathcal{C}_k) \pi_k}, \end{aligned} \quad (14)$$

where $|\mathcal{C}|$ is the number of classes which is two here. The $f_1(x)$ and π_1 are the *likelihood (class conditional)* and *prior* probabilities, respectively, and the denominator is the marginal probability.

Therefore, Eq. (13) becomes:

$$\begin{aligned} &\frac{f_1(x) \pi_1}{\sum_{i=1}^{|\mathcal{C}|} \mathbb{P}(X = x | x \in \mathcal{C}_i) \pi_i} \\ &\stackrel{\text{set}}{=} \frac{f_2(x) \pi_2}{\sum_{i=1}^{|\mathcal{C}|} \mathbb{P}(X = x | x \in \mathcal{C}_i) \pi_i} \\ \implies f_1(x) \pi_1 &= f_2(x) \pi_2. \end{aligned} \quad (15)$$

Now let us think of data as *multivariate* data with dimensionality d . The PDF for multivariate Gaussian distribution, $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is:

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}} \exp \left(-\frac{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{2} \right), \quad (16)$$

where $\mathbf{x} \in \mathbb{R}^d$, $\boldsymbol{\mu} \in \mathbb{R}^d$ is the mean, $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ is the covariance matrix, and $|\cdot|$ is the determinant of matrix. The $\pi \approx 3.14$ in this equation should not be confused with the π_k (prior) in Eq. (12) or (15). Therefore, the Eq. (12) or (15) becomes:

$$\begin{aligned} &\frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}_1|}} \exp \left(-\frac{(\mathbf{x} - \boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}_1^{-1} (\mathbf{x} - \boldsymbol{\mu}_1)}{2} \right) \pi_1 \\ &= \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}_2|}} \exp \left(-\frac{(\mathbf{x} - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_2^{-1} (\mathbf{x} - \boldsymbol{\mu}_2)}{2} \right) \pi_2, \end{aligned} \quad (17)$$

where the distributions of the first and second class are $\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$, respectively.

3. Linear Discriminant Analysis for Binary Classification

In Linear Discriminant Analysis (LDA), we assume that the two classes have equal covariance matrices:

$$\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}. \quad (18)$$